

CS & IT ENGINEERING



Operating System

CPU Scheduling

Lecture -3



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Recap of Previous Lecture



Topic

LF, LRTF Scheduling

Topic

HRRN Algorithms

Topic

Priority Based Scheduling

Topic

Round Robin Scheduling

Topics to be Covered



Topic

Multilevel Queue Scheduling

Topic

Multilevel Feedback Queue Scheduling

Topic

Questions on Scheduling



Topic : Round Robin (RR)

Scheduling Criteria: Arrival time and Quantum time

Type of Algorithm: preemptive

if 2 processes arrive together $\Rightarrow$ Add the process of smaller id first in queue.
(equal AT)

if a process arrives at the same time when the current running process is preempted $\Rightarrow$ Add the new process first and preempted process in last in queue.



Topic : Round Robin (RR)

$$Q = 2$$

Process	Arrival Time	Burst Time
P1	0	3
P2	0	6
P3	0	4
P4	0	5

no. of context switches

$$= \left\lceil \frac{3}{2} \right\rceil + \left\lceil \frac{6}{2} \right\rceil + \left\lceil \frac{4}{2} \right\rceil + \left\lceil \frac{5}{2} \right\rceil - 1$$

Time	Ready Queue
0	P1 , P2, P3, P4
2	P2 , P3, P4, P1
4	P3 , P4, P1, P2
6	P4 , P1, P2, P3
8	P1 , P2, P3, P4
9	P2, P3 , P4



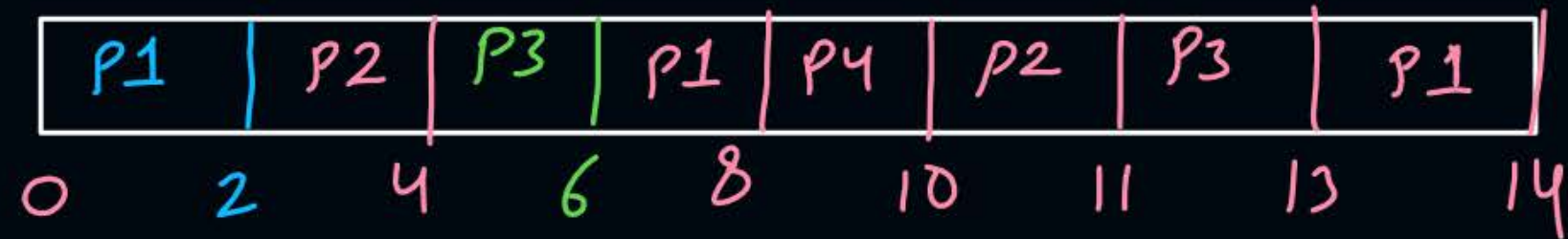
no. of context switches = 9

Ques-2)

Pid	AT	BT
P1	0	5
P2	1	3
P3	2	4
P4	3	2

Time	Ready Queue
0	P1
2	P2 , P3, P1
4	P3 , P1, P4, P2
6	P1, P4, P2, P3 X X X X

Q = 2



no. of context switches = 7



Topic : Round Robin (RR)

Q = 2

Process	Arrival Time	Burst Time
P1	0	6 3
P2	1	5 3
P3	2	4 2
P4	3	3 2
P5	4	2 1
P6	5	4 2

Time	Ready Queue
0	P1
2	P2, P3, P1
4	P3, P1, P4, P5, P2
6	P1, P4, P5, P2, P6, P3 X X X X X X

$\frac{2}{13} - 1 = 12$ ← context switches

p1	p2	p3	p1	p4	p5	p2	p6	p3	p1	p4	p2	p6	
0	2	4	6	8	10	12	14	16	18	20	21	22	24



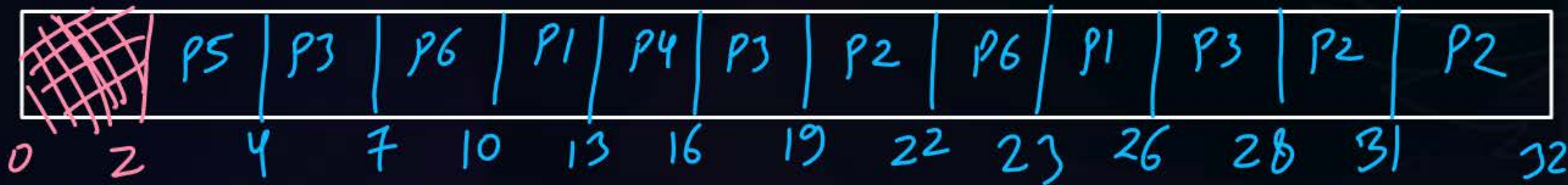
Topic : Round Robin (RR)

$Q = 3$

Process	Arrival Time	Burst Time
P1	5	6
P2	8	7
P3	3	8
P4	6	3
P5	2	2
P6	4	4

Time	R. Q.
2	P5
4	P3, P6
7	P6, P1, P4, P3
10	P1, P4, P3, P2, P6

$12 - 1 = 11 \leftarrow$ no. of context switches





Topic : Round Robin (RR)

Q = 3

H.W.



Process	Arrival Time	Burst Time
P1	0	12
P2	0	5
P3	3	9
P4	5	6
P5	2	8
P6	4	2
P7	1	7



Topic : What Should Be the Quantum Value?

very-very small

CPU efficiency ≈ 0

CPU spends more time on context switch than process execution

$\approx$ small

Interactive

large

less interactive

very-very large

RR degrades to FCFS



Topic : RR



Advantages:

1. All processes execute one by one, so no starvation
2. Better interactive ~~ness~~ *ness*
3. Burst time is not required to be known in advance $\Rightarrow$ *Practically implemented*

Disadvantages:

1. Average waiting time and turnaround time ~~is~~ *are* more
2. Can degrade to FCFS

[MCQ]

#Q. If the time-slice used in the round-robin scheduling policy is more than the maximum time required to execute any process, then the policy will?

- ☒ **A** Degenerate to shortest job first
- ☐ **B** Degenerate to priority scheduling
- ☒ **C** Degenerate to first come first serve
- ☐ **D** None of the above

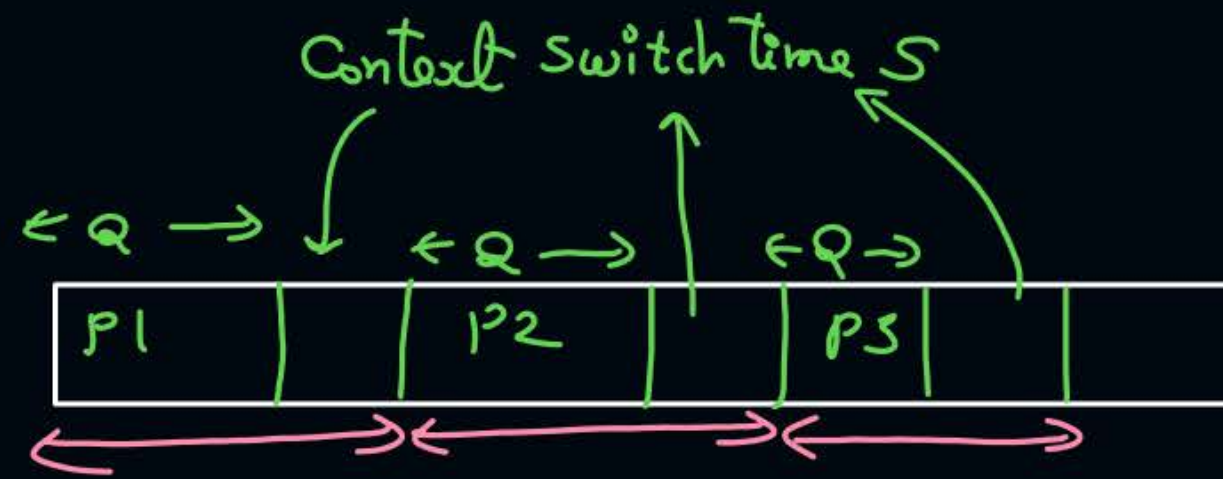


H.W.

#Q. A scheduling algorithm assigns priority proportional to the waiting time of a process. Every process starts with priority zero (the lowest priority). The scheduler re-evaluates the process priorities every T time units and decides the next process to schedule. Which one of the following is TRUE if the processes have no I/O operations and all arrive at time zero?

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- A** This algorithm is equivalent to the first-come-first-serve algorithm
- B** This algorithm is equivalent to the round-robin algorithm.
- C** This algorithm is equivalent to the shortest-job-first algorithm
- D** This algorithm is equivalent to the shortest-remaining-time-first algorithm



$$\text{CPU efficiency} = \frac{Q}{Q + S}$$



Topic : Multilevel Queue Scheduling

↓
There are multiple ready queue.
and processes are kept in queues as per their behaviour
for each queue a diff. algo is used.



Topic : Multilevel Queue Scheduling

3 queues

System Processes

→ RR $Q=2$

Foreground Processes

→ RR $Q=4$

Background Processes

→ FCFS



Topic : Multilevel Queue Scheduling

1. Fixed priority preemptive scheduling method $\Rightarrow$ *starvation*
2. Time slicing



Topic : Multilevel Queue Scheduling

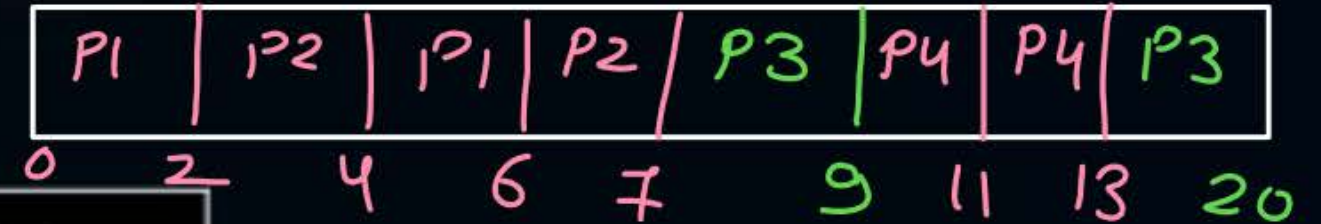
fixed priority preemptive

Queue 1: RR with Q=2
(Higher)

Q1: ~~P1, P2~~, P4

Queue 2: FCFS
(Lower)

Q2: P3



Process	Arrival Time	Burst Time	Queue
P1	0	4 2	1
P2	0	3 1	1
P3	0	9 7	2
P4	9	4 7	1



Topic : Multilevel Queue Scheduling

H.W.

Queue 1: RR with $Q=3$
Higher

Queue 2: FCFS

Process	Arrival Time	Burst Time	Queue
P1	0	3	1
P2	0	3	1
P3	2	8	2
P4	10	4	1
P5	11	6	2
P6	11	3	1
P7	19	2	1
P8	13	5	2



Topic : Multilevel Queue Scheduling

Disadvantages:

1. Some processes may starve for CPU if some higher priority queues are never becoming empty
2. It is inflexible in nature. *→ processes can not be shifted b/w queue.*



Topic : Multilevel Feedback Queue Scheduling

Processes can be moved to higher or lower priority queue.



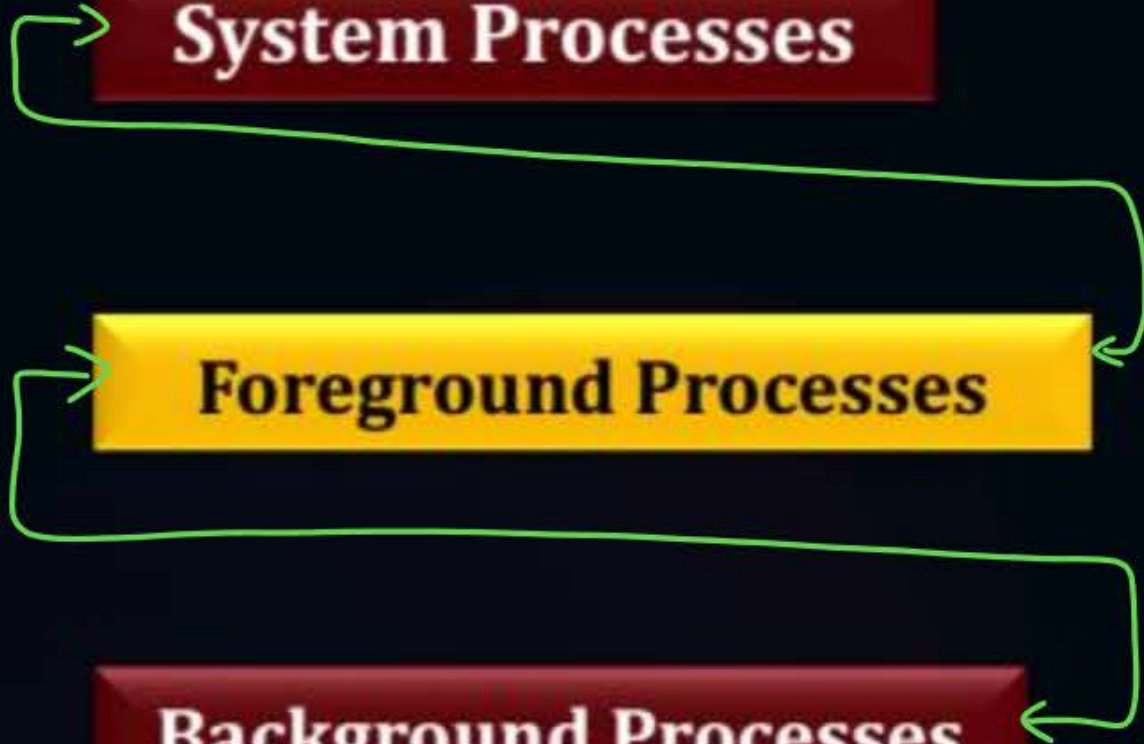


Topic : Multilevel Feedback Queue Scheduling

System Processes

Foreground Processes

Background Processes





Topic : Multilevel Feedback Queue Scheduling

Disadvantage:

1. Some processes may starve for CPU if some higher priority queues are never becoming empty.

Advantage:

1. Flexible



#Q. Consider the following set of processes:

Process	Arrival Time	Burst Time
P1	0	10ms
P2	0	29ms
P3	0	3ms
P4	0	7ms
P5	0	12ms

FCFS = 28

SJF = 13

SRTF = 13

RR = 23

Calculate average waiting time for:

FCFS, Non-preemptive SJF, SRTF and Round-robin (quantum = 10ms)



2 mins Summary

Topic

Multilevel Queue Scheduling

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Multilevel Feedback Queue Scheduling

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Questions on Scheduling





Happy Learning

THANK - YOU

